

Assignment 6: Research Workflow Automation with GitHub

Step 1: Install Required Libraries

```
PS C:\Users\hanmant\Desktop\sdAssignment\Assignment6> python -m pip install streamlit pillow numpy
Collecting streamlit
  Downloading streamlit-1.55.0-py3-none-any.whl.metadata (9.8 kB)
Collecting pillow
  Downloading pillow-12.1.1-cp314-cp314-win_amd64.whl.metadata (9.0 kB)
Collecting numpy
  Downloading numpy-2.4.3-cp314-cp314-win_amd64.whl.metadata (6.6 kB)
Collecting altair!=5.4.0,!=5.4.1,<7,>=4.0 (from streamlit)
  Downloading altair-6.0.0-py3-none-any.whl.metadata (11 kB)
```

Step 2: Create Python File(skin_cancer_app.py)

```
skin_cancer_app.py 3 X
C: > Users > hanmant > Desktop > sdAssignment > Assignment6 > skin_cancer_app.py > ...
1  import streamlit as st
2  from PIL import Image
3  import numpy as np
4  import random
5
6  st.title("Skin Cancer Detection Dashboard")
7
8  st.write("Upload a skin lesion image to analyze.")
9
10 uploaded_file = st.file_uploader("Upload Image", type=["jpg","png","jpeg"])
11
12 classes = [
13     "Melanoma",
14     "Melanocytic Nevi",
15     "Basal Cell Carcinoma",
16     "Actinic Keratosis",
17     "Benign Keratosis"
18 ]
19
20 if uploaded_file is not None:
21
22     image = Image.open(uploaded_file)
23     st.image(image, caption="Uploaded Image", use_column_width=True)
24
25     # Demo prediction (random for assignment)
26     prediction = random.choice(classes)
27     confidence = random.randint(80,99)
28
29     st.subheader("Prediction Result")
30
31     st.write("Predicted Class:", prediction)
32     st.write("Confidence:", confidence,"%")
33
34     if prediction == "Melanoma":
35         st.error("High Risk Skin Cancer")
36     else:
37         st.success("Low Risk Lesion")
```

Step 3: Run Streamlit Application

```
PS C:\Users\hanmant\Desktop\sdAssignment\Assignment6> python -m streamlit run skin_cancer_app.py

🔥 Welcome to Streamlit!

If you'd like to receive helpful onboarding emails, news, offers, promotions,
and the occasional swag, please enter your email address below. Otherwise,
leave this field blank.

Email: skincancer_dummyweb@gmail.com

You can find our privacy policy at https://streamlit.io/privacy-policy

Summary:
- This open source library collects usage statistics.
- We cannot see and do not store information contained inside Streamlit apps,
  such as text, charts, images, etc.
- Telemetry data is stored in servers in the United States.
- If you'd like to opt out, add the following to %userprofile%\.streamlit/config.toml,
  creating that file if necessary:

[browser]
gatherUsageStats = false

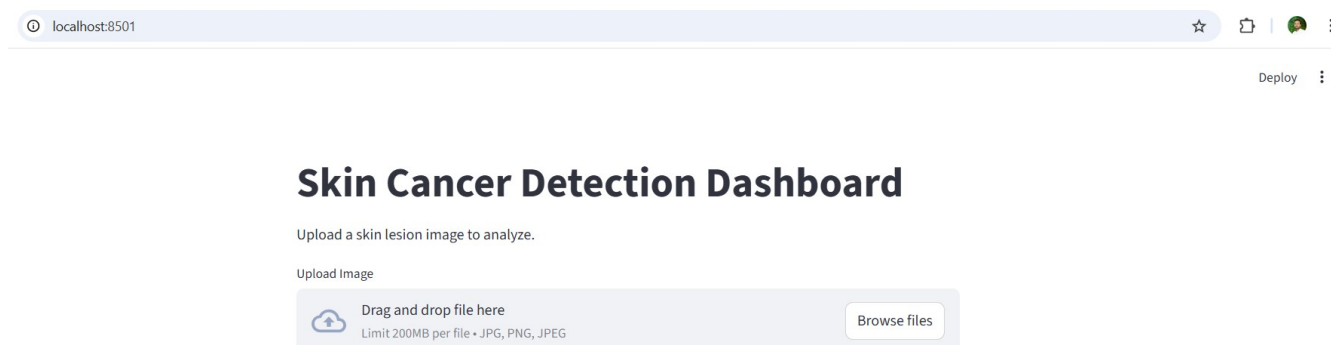
You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://10.28.54.74:8501

2026-03-16 17:54:58.059 The 'use_column_width' parameter has been deprecated and will be removed in
instead.
```

Step 4: Dashboard Look

(Open the browser and go to <http://localhost:8501>)



Step 5: View Prediction Result

Skin Cancer Detection Dashboard

Upload a skin lesion image to analyze.

Upload Image



Drag and drop file here

Limit 200MB per file • JPG, PNG, JPEG

Browse files



ISIC_0029334.jpg 268.4KB



The `use_column_width` parameter has been deprecated and will be removed in a future release. Please utilize the `width` parameter instead.



Uploaded Image

Prediction Result

Predicted Class: Basal Cell Carcinoma

Confidence: 94 %

Low Risk Lesion